PHYSICS I · LECTURE 1

Getting the Numbers Right

Course overview · Units · Scientific notation · Significant figures

Textbook section 1.8



What This Course Covers

01 **Motion** *“kinematics”*

Position, velocity, acceleration.

02 **Forces** *“mechanics”*

Pushes and pulls, and what they do.

03 **Energy & Momentum**

A different way to solve mechanics problems

04 **Oscillations**

Vibrations, pendulums, springs.

I Want You to Succeed

Physics is hard.

The hard part isn’t the arithmetic. It’s translating the physical world into math.

That’s a skill. Skills are trainable.

My job in this room

- Guide your learning
- Clear up confusion
- Show you why it’s fun

Build a Rhythm Around Lecture

BEFORE

Skim the week’s sections.

DURING

Don’t just transcribe. Listen and engage

AFTER

Read what we covered — same day if you can.

Read the book. I won’t cover everything.

Lecture is a guide to what matters.

But Mostly: Solve Problems

15 **minutes a day**

beats three hours the night before.

TUE **problem session**

We practice solving problems together!

Why problems, not re-reading?

- Physics is a skill, not a list of facts.
- You find your gaps by hitting them.
- Reading a solution isn’t solving it.
- Exams ask you to produce, not recognize.

SECTION 1.8

Units & Scientific Notation

Attaching numbers to the physical world.



A Unit Connects a Number to a Thing

“5” is not an answer. “5 meters” is.

- SI (metric) units are the standard in science.
- Three base units describe all of motion.
- Everything else is a combination of them.

BASE UNITS

Time	**second**	**s**
Length	**meter**	**m**
Mass	**kilogram**	**kg**

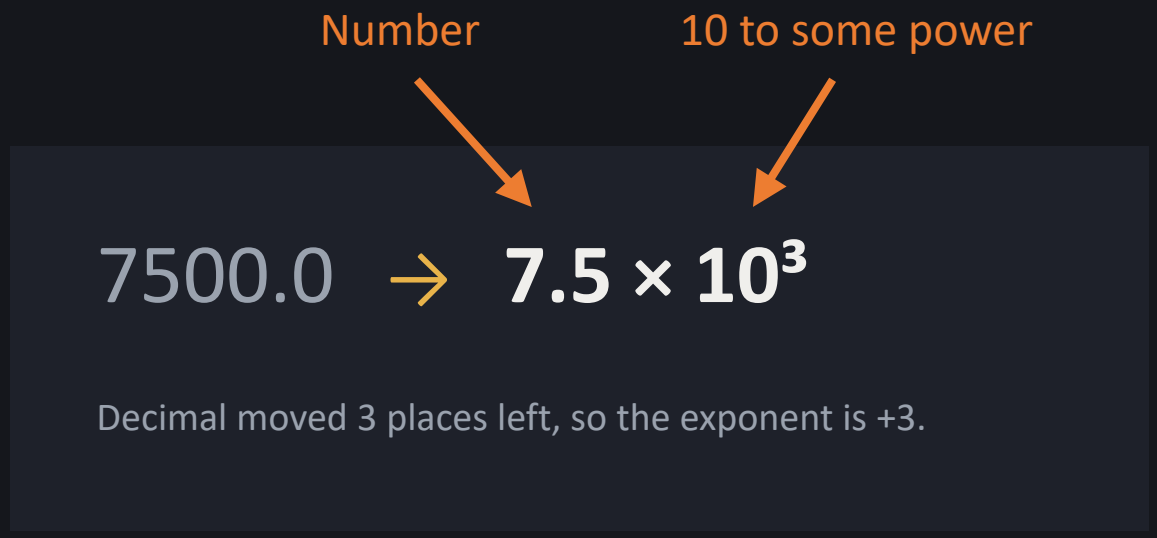
DERIVED FROM THEM

Velocity	Acceleration	Force
m/s	**m/s^2**	**$kg \cdot m/s^2$**

Scientific Notation

A number between 1 and 10, times a power of ten.

- Used everywhere in science.
- Makes huge and tiny numbers writable.
- Makes math/calculator easier



THE EXPONENT FOLLOWS THE DECIMAL

Decimal moves left	**exponent up**
Decimal moves right	**exponent down**

Examples

- Distance of Earth to the Sun:

  93,000,000 miles

- Size of red blood cell:

  0.000,167 inches across



Prefixes Are Named Powers of Ten

Capitalization matters: m is milli, M is mega.

BIGGER THAN ONE

tera 10^{12}	**T**	**giga** 10^{9}	**G**
mega 10^{6}	**M**	**kilo** 10^{3}	**k**

SMALLER THAN ONE

milli 10^{-3}	**m**	**micro** 10^{-6}	**μ**
nano 10^{-9}	**n**	**pico** 10^{-12}	**p**

- We use prefixes in powers of 10^3

- Giga-byte (GB)
- Milli-gram (mg)

The full list runs to quetta (10^{30}) and quecto (10^{-30}). https://en.wikipedia.org/wiki/Metric_prefix)

Unit Conversions

Multiply by one — carefully chosen.

1 **Write the factor as a ratio equal to 1.**

1 mi = 1.609 km

2 **Multiply your value by that ratio.**

Multiplying by 1 changes nothing.

3 **Check that the unwanted unit cancels.**

ON THE BOARD

Earth’s radius: 3,959 mi

in kilometers?

ON THE BOARD

Walking speed: 3 mph

in meters per second?

SECTION 1.8, CONTINUED

Significant Figures

How many digits you write is itself a claim.

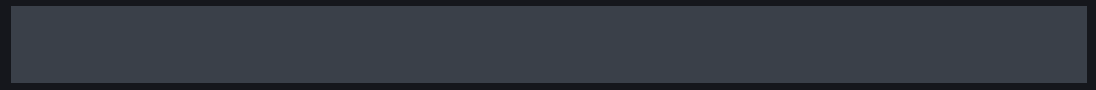


Digits Carry Information

A significant figure is a digit you actually know.

6 m

1 sig fig



5.5 – 6.5 m

6.2 m

2 sig figs



6.15 – 6.25 m

6.213 m

4 sig figs



6.2125 – 6.2135 m

Bar length = how tightly the value is pinned down (not to scale).

Who decides?

Your instrument does. A meter stick earns you 6.2 m, not 6.213 m.

The Rules

Tactics Box 1.5.

× ÷ Multiply, divide, roots

Keep the fewest sig figs of any input.

+ − Add, subtract

Keep the fewest decimal places.

∞ Exact numbers

Counts and definitions never limit sig figs.

… Intermediate steps

Carry extra digits, round once at the end.

One Last Thing

Don’t hand me your calculator’s opinion.

DON’T WRITE

1 / 7.0 = 0.1428571423

Ten digits from a two-digit input.

DO WRITE

1 / 7.0 = 0.14

Two sig figs in, two out.

Next time: position, velocity, and acceleration.

